Research Evolution

Project 1 | Frozen record supporting the final recommendation

Edition A status

This record preserves the research evolution without revising or relabeling later work as the original Edition A. It documents Edition B and the evidence supporting its final architecture.

Edition B decision question

Should a buy-side investment committee with no current Tesla position initiate, watch/defer, or not initiate TSLA at September 30, 2026, based on a valuation range with explicit conditions and reversal triggers?

A-to-B evolution

Area	Edition B conclusion	Evidence / reason
Decision	Do not initiate at $354.81; retain TSLA on a watch list.	Frozen DCF and qualified GM references are far below the market reference.
Data	Dated closes cross-checked; risk-free rate corrected to 5.29%; bridge uses FY2025 filed cash, investments, debt, and leases.	Federal Reserve H.15 and Tesla 2025 Form 10-K reconciliation.
Model	Five-year linked FCFF/WACC DCF, EV-to-equity bridge, sensitivity, reverse DCF, and operating-cash-reserve sensitivity.	Replaces lab-stage FCFE work with an enterprise-value framework.
Uncertainty	Terminal value is 99.0% of EV; GM is a qualified one-peer reference; reported liquidity may not all be excess cash.	These limitations prevent a point-price conclusion.

What changed and why

The final system moves the work from disconnected lab-stage valuation toward a linked FCFF/WACC engine that begins with a five-year three-statement projection and ends with a stated enterprise-to-equity bridge. The dated-input review corrected the preliminary 5.26% risk-free rate to 5.29%, resulting in a 14.46% WACC and $21.21 base DCF value per diluted share.

The recommendation is conditional rather than a claim that the market is wrong. At the frozen explicit cash flows, matching the market reference requires approximately 14.06% terminal growth, near the 14.46% WACC and outside a credible perpetuity assumption. GM is used as corroboration only: its ICE-heavy mix, dealer model, and GM Financial business make it an imperfect economic comparable.

AI-use evolution

Codex supported project structuring, code assistance, research leads, and explanatory drafting. The dated risk-free-rate result was independently checked against the Federal Reserve H.15 series and corrected before the model was rerun. No AI-generated judgment was accepted without source evidence; the locked changed-input prediction was authored by the student without AI assistance.